Budget Bites Project Report

AjxOm undergraduate Project One
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Budget Bites is a device-local lunch notebook. Students save meal estimates, record actual purchase amounts, correct receipt history and compare a transparent history-based estimate with a lunch budget. The application works without an account or a network service.

Implemented scope

Screen role	Delivered behavior
Catalog	Saved meal cards, search/cuisine/favorites filters, add/edit/delete and budget comparison
Meal editor	Validated meal name, venue note, item labels and USD prices
Detail and activity	Start, draft progress, explicit save/finish, over-budget confirmation, reset, local helper
History	A.J. card/list layout integrated with receipt CRUD, date/meal filters and spending total
Settings	Budget slider and dark mode stored on-device

Product and data choices

Two SQLite entities, Meal and PurchaseLog, keep catalog estimates separate from historical actual purchases. Favorites is a flag on Meal. Prices use integer cents. The starter meals and QA purchase fixtures are fictional. No current restaurant prices, delivery, payment, remote inference or cloud database are implied.

The user confirmed instructor-team approval before this implementation continued. The signed statement and submitted proposal accompany the private course package.

Architecture and integration

A.J. authored the initial Meal model, SQLite methods and standalone History page on the data branch. Om's integration branch preserved those commits, corrected incompatible types/navigation and connected the final History screen to PurchaseLog. Both histories are combined through a merge rather than rewritten or reassigned.

Class responsibilities

Component	Responsibility
BudgetBitesApp	App theme and lifetime of AppStore/database
HomeScreen	Tabs, meal actions and navigation callbacks
MealHistory	Receipt list, meal/date filtering, total and Card/ListTile presentation
MealEditor and PurchaseEditor	Validated transient forms and pending-save state
AppStore	Shared meals/purchases, refresh and preferences
MealDatabase	Connection owner, constraints, migration and parameterized CRUD
Domain helpers	Exact money parsing and pure Lunch Lens calculation

Integrity and lifecycle

PurchaseLog.mealId references Meal.id with restricted deletion. Linked receipts must be removed or reassigned before the parent can be deleted. Receipt deletion requires confirmation. A stable unique draft token makes repeat/retry saves idempotent. Updates refresh both totals and estimates. Editing the catalog estimate never rewrites a prior actual amount.

Version 2 converts the earlier REAL dollar fields into integer cents while preserving IDs. A device fixture verifies that ID 17 survives with a 650-cent total. Forms dispose controllers and guard asynchronous UI updates after navigation. Reset clears only the unsaved draft. SharedPreferences stores budget and dark mode; active-draft process restoration is outside scope.

Integration problems resolved

The original table had a trailing SQL comma and REAL/int price conflicts. The standalone History page indexed an integer ID as a string and invoked a private State class instead of navigating. Integration keeps the shared database, uses valid initials and routes receipt edits to PurchaseEditor. History deletes child receipts, not catalog meals.

Local helper and evaluation

Lunch Lens is a local rule-based moving-average estimate. It uses the three latest purchases of the selected meal strictly before the chosen prediction time. It shows the inputs, date range, estimate and last-price baseline. Not now dismisses it. It never silently fills the actual amount or saves a purchase.

Case	Expected and actual result	Baseline / reference
H1 normal: 450,600,750 cents	600 cents; pass	750 / 650
H2: only 450,600	Insufficient-history feedback; pass	Not scored
H3: 450,-1,750	Invalid-data exception; pass	Not scored
H4: 100,100,101	100 cents; pass	101 / 100
H5: 500,2000,500	1000 cents; pass	500 / 500

These are designed fictional development fixtures, not real spending or held-out evidence. Across the three valid comparisons, mean absolute error is 183.33 cents for the mean and 33.67 cents for last price, n=3. Empty and invalid cases assess feedback rather than numeric prediction accuracy.

Critical thinking and limitations

Meal identity, amount and timestamp are relevant inputs. Sensitive personal information is unnecessary. Three recent records are easy to explain but can mix promotions and extras. A median-of-three is a reasonable alternative. The outlier case demonstrates that the baseline can win. We retain the brief's moving mean with the limitation disclosed and user control preserved, without claiming broad accuracy or causal benefit.

The implementation excludes future/reference rows, uses deterministic timestamp/ID ties, rounds to integer cents and recomputes after corrections. AI_FEATURE.md records rationale, alternative, cases, metrics, counterexample and user-control/offline evidence.

Presentation evidence

For R4, show three saved inputs, the suggestion and explanation, dismissal, the insufficient-history state, then the H1-H5 table and outlier. Explain that the over-budget warning is ordinary validation and does not itself satisfy the required helper.

Validation and completion status

Check	Recorded result
Analyzer and unit/widget tests	Final logs identify the exact integration result. Suite includes 17 logic/widget checks, including three new History checks.
Two emulator profiles	API 37 Pixel_8 and a separate API 37 emulator with a larger display. Actual SQLite CRUD, duplicate-token, FK and migration tests.
Release and restart	Release APK built. Prior release offline force-stop/relaunch checks retained genuine screenshots; final integrated release receives its own build/install verification.
Responsive behavior	Narrow History with enlarged text has a targeted automated check. Earlier landscape/large-text screenshots remain dated development evidence.

Use evidence/final-analyze.txt, final-unit-widget-tests.txt, final-phone-integration.txt, final-large-integration.txt and final-release-build.txt as the authoritative current logs. Older October 26 logs/screenshots are retained as historical evidence and are not relabeled as October 30 results.

Ownership and learning

A.J. owns the original data foundation and History-page contribution. Om is responsible for frontend integration, navigation, corrected persistence wiring, helper, tests and packaging.

Course references: codd.cs.gsu.edu/~lhenry23/mad/Projects/Project1/ver2/assessment/grading-rubric.html and submission/About_Submission.html. Repository: github.com/Omswaroopreddyudumula/ajxom-budget-bites.